

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Alibaba B2B E-Commerce Database Management System

Case Study: JBL Bluetooth Earbuds Wholesale Sourcing (Supplier = Seller entity)

1. Database Design

1.1 Supplier (Seller) Table

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Location	Not Null
MOQ_Policy	-
Certification_Status	Check Key
Contact_No	Unique Key
Email	Unique Key

1.2 Inventory Table

Attribute	Key Type
Inventory_ID	Primary Key
Supplier_ID	Foreign Key
Product_ID	Foreign Key
Stock_Quantity	Default, Check Key
Stock_Status	Default, Check Key
Last_Updated	Not Null

2. Primary and Foreign Key Relationship

Supplier → Inventory

- Supplier.Supplier_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Supplier_ID → Foreign Key referencing Supplier.Supplier_ID
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-Many (1:M) — one supplier can have multiple inventory records.

3. Table Creation

3.1 Create Supplier Table

```
SQL> CREATE TABLE Supplier (  
2     Supplier_ID NUMBER PRIMARY KEY,  
3     Supplier_Name VARCHAR2(100) NOT NULL,  
4     Location VARCHAR2(100) NOT NULL,  
5     MOQ_Policy VARCHAR2(100),  
6     Certification_Status VARCHAR2(30)  
7     CHECK (Certification_Status IN ('Verified', 'Pending')),  
8     Contact_No VARCHAR2(20) UNIQUE,  
9     Email VARCHAR2(100) UNIQUE  
10 );
```

Table created.

The Supplier table is created with Supplier_ID as the primary key. It stores supplier details such as supplier name, location, MOQ policy, certification status, contact number, and email. Contact_No and Email are unique, and Certification_Status is restricted by a CHECK constraint.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Supplier_ID NUMBER NOT NULL,  
4     Product_ID NUMBER NOT NULL,  
5     Stock_Quantity NUMBER DEFAULT 0  
6     CHECK (Stock_Quantity >= 0),  
7     Stock_Status VARCHAR2(20) DEFAULT 'In Stock'  
8     CHECK (Stock_Status IN ('In Stock', 'Out of Stock')),  
9     Last_Updated DATE NOT NULL,  
10    CONSTRAINT fk_inv_supplier  
11        FOREIGN KEY (Supplier_ID)  
12        REFERENCES Supplier(Supplier_ID),  
13    CONSTRAINT fk_inv_product  
14        FOREIGN KEY (Product_ID)  
15        REFERENCES Product(Product_ID)  
16 );
```

Table created.

The Inventory table is created with Inventory_ID as the primary key. Supplier_ID and Product_ID are foreign keys that establish relationships with the Supplier and the existing Product table. Stock_Quantity defaults to 0, Stock_Status defaults to 'In Stock', and CHECK constraints prevent negative stock and invalid status values.

4. Data Insertion

4.1 Supplier Records

Five supplier records are inserted into the Supplier table, followed by a COMMIT to save the transaction.

```
SQL> INSERT INTO Supplier
  2 VALUES (1, 'Shenzhen SoundWave Electronics', 'Shenzhen, China',
  3           'MOQ 500 pcs', 'Verified', '+8613800138001',
  4           'sales@soundwave-sz.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (2, 'Guangzhou AudioTech Co.', 'Guangzhou, China',
  3           'MOQ 300 pcs', 'Verified', '+8613800138002',
  4           'info@audiotech-gz.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (3, 'Dongguan BassPro Trading', 'Dongguan, China',
  3           'MOQ 1000 pcs', 'Pending', '+8613800138003',
  4           'contact@basspro-dg.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (4, 'Huizhou EarPod Manufacturing', 'Huizhou, China',
  3           'MOQ 200 pcs', 'Verified', '+8613800138004',
  4           'export@earpod-hz.com');

1 row created.
```

```
SQL> INSERT INTO Supplier
  2 VALUES (5, 'Foshan Wireless Audio Ltd.', 'Foshan, China',
  3           'MOQ 500 pcs', 'Verified', '+8613800138005',
  4           'hello@wirelessaudio-fs.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (6, 'Ningbo SoundTech Ltd.', 'Ningbo, China',
  3           'MOQ 400 pcs', 'Verified', '+8613800138006',
  4           'sales@soundtech-nb.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (7, 'Shanghai AudioPro Industries', 'Shanghai, China',
  3           'MOQ 600 pcs', 'Pending', '+8613800138007',
  4           'info@audiopro-sh.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2 VALUES (8, 'Xiamen Wireless Hub', 'Xiamen, China',
  3           'MOQ 250 pcs', 'Verified', '+8613800138008',
  4           'contact@wirelesshub-xm.com');

1 row created.
```

```
SQL>
SQL> INSERT INTO Supplier
  2  VALUES (9, 'Wenzhou Smart Audio Co.', 'Wenzhou, China',
  3           'MOQ 350 pcs', 'Verified', '+8613800138009',
  4           'sales@smartaudio-wz.com');

1 row created.

SQL>
SQL> INSERT INTO Supplier
  2  VALUES (10, 'Hangzhou Bluetooth Tech', 'Hangzhou, China',
  3           'MOQ 700 pcs', 'Pending', '+8613800138010',
  4           'export@bluetooth-hz.com');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

4.2 Inventory Records

Inventory records are inserted with supplier IDs, product IDs, stock quantities, stock status, and last updated dates. A COMMIT saves the transaction.

```
SQL> INSERT INTO Inventory
  2 VALUES (101, 1, 101, 5000, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (102, 2, 102, 2500, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (103, 3, 103, 0, 'Out of Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (104, 4, 104, 1500, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (105, 5, 105, 3500, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```
SQL>
SQL> INSERT INTO Inventory
  2 VALUES (106, 6, 106, 0, 'Out of Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (107, 7, 107, 4200, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (108, 8, 108, 1800, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (109, 9, 109, 2750, 'In Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (110, 10, 110, 0, 'Out of Stock',
  3         TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.
```

Data Retrieval

5.1 Display Supplier Records

All supplier records are retrieved to verify successful insertion.

```
SQL> SELECT * FROM Supplier;

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MOQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
1

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MOQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
Shenzhen SoundWave Electronics

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
```

```
-----
SUPPLIER_NAME
-----
LOCATION
-----
MOQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
Shenzhen, China

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MOQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
MOQ 500 pcs

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MOQ_POLICY
-----
```

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Verified +8613800138001

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

sales@soundwave-sz.com

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

2

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Guangzhou AudioTech Co.

SUPPLIER_ID

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Guangzhou, China

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

MOQ 300 pcs

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Verified +8613800138002

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

info@audiotech-gz.com

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

3

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Dongguan BassPro Trading

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

```
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
Dongguan, China
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
MQQ 1000 pcs
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
```

```
-----
Pending                    +8613800138003
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
contact@basspro-dg.com
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
```

```
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
4
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
Huizhou EarPod Manufacturing
-----
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MQQ_POLICY
-----
```

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Huizhou, China

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

MOQ 200 pcs

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Verified +8613800138004

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

export@earpod-hz.com

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

SUPPLIER_ID

SUPPLIER_NAME

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

5

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Foshan Wireless Audio Ltd.

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Foshan, China

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

MOQ 500 pcs

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Verified +8613800138005

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

hello@wirelessaudio-fs.com

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

SUPPLIER_ID

SUPPLIER_NAME

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

6

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Ningbo SoundTech Ltd.

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

MOQ 400 pcs

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Verified +8613800138006

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

sales@soundtech-nb.com

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

7

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Shanghai AudioPro Industries

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

CERTIFICATION_STATUS CONTACT_NO

EMAIL

Shanghai, China

SUPPLIER_ID

SUPPLIER_NAME

LOCATION

MOQ_POLICY

```

CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
MDQ 600 pcs
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MDQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
Pending                    +8613800138007
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MDQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
info@audiopro-sh.com

```

```

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MDQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
export@bluetooth-hz.com
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
LOCATION
-----
MDQ_POLICY
-----
CERTIFICATION_STATUS      CONTACT_NO
-----
EMAIL
-----
10 rows selected.

```

5.2 Display Inventory Records

All inventory records are retrieved to verify the stock information.

```

SQL> SELECT * FROM Inventory;

INVENTORY_ID SUPPLIER_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
LAST_UPDA
-----
      101          1       101          5000 In Stock
19-SEP-26
      102          2       102          2500 In Stock
19-SEP-26
      103          3       103              0 Out of Stock
19-SEP-26

INVENTORY_ID SUPPLIER_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
LAST_UPDA
-----
      104          4       104          1500 In Stock
19-SEP-26
      105          5       105          3500 In Stock
19-SEP-26
      106          6       106              0 Out of Stock
19-SEP-26

```

```
INVENTORY_ID SUPPLIER_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
LAST_UPDA
-----
      107          7       107       4200 In Stock
19-SEP-26

      108          8       108       1800 In Stock
19-SEP-26

      109          9       109       2750 In Stock
19-SEP-26

INVENTORY_ID SUPPLIER_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
LAST_UPDA
-----
      110         10       110           0 Out of Stock
19-SEP-26

10 rows selected.
```

6. Supplier Product Information

A JOIN query is used to display the supplier ID, supplier name, product name, stock quantity, stock status, and last updated date.

```
SQL> SELECT
  2     s.Supplier_ID,
  3     s.Supplier_Name,
  4     i.Inventory_ID,
  5     i.Product_ID,
  6     i.Stock_Quantity,
  7     i.Stock_Status,
  8     i.Last_Updated
  9 FROM Supplier s
 10 JOIN Inventory i
 11 ON s.Supplier_ID = i.Supplier_ID
 12 ORDER BY s.Supplier_Name;
```

```
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
3
Dongguan BassPro Trading
103      103              0 Out of Stock    19-SEP-26

5
Foshan Wireless Audio Ltd.
105      105             3500 In Stock     19-SEP-26

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
2
Guangzhou AudioTech Co.
```

```
2
Guangzhou AudioTech Co.
102      102             2500 In Stock     19-SEP-26

10
Hangzhou Bluetooth Tech

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
110      110              0 Out of Stock    19-SEP-26

4
Huizhou EarPod Manufacturing
104      104             1500 In Stock     19-SEP-26

6

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
Ningbo SoundTech Ltd.
106      106              0 Out of Stock    19-SEP-26

7
Shanghai AudioPro Industries
107      107             4200 In Stock     19-SEP-26
```

```
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
1
Shenzhen SoundWave Electronics
101      101             5000 In Stock     19-SEP-26

9
Wenzhou Smart Audio Co.
109      109             2750 In Stock     19-SEP-26

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
INVENTORY_ID PRODUCT_ID STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
-----
8
Xiamen Wireless Hub
108      108             1800 In Stock     19-SEP-26

10 rows selected.
```

10 rows selected.

7. Product Availability

7.1 Available Products

Products with stock available are displayed using the Stock_Status field.

```
SQL> SELECT
  2     Product_ID,
  3     Stock_Quantity,
  4     Stock_Status
  5 FROM Inventory
  6 WHERE Stock_Status = 'In Stock';

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
      101         5000 In Stock
      102         2500 In Stock
      104         1500 In Stock
      105         3500 In Stock
      107         4200 In Stock
      108         1800 In Stock
      109         2750 In Stock

7 rows selected.
```

7.2 Unavailable Products

Products with zero stock or Out of Stock status are displayed separately.

```
SQL> SELECT
  2   Product_ID,
  3   Stock_Quantity,
  4   Stock_Status
  5 FROM Inventory
  6 WHERE Stock_Status = 'Out of Stock';
```

PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
103	0	Out of Stock
106	0	Out of Stock
110	0	Out of Stock

8. Inventory Update

Demonstrate updating stock information: two out-of-stock products (Inventory_ID 110 and 102) are restocked, and their status and last updated date are refreshed. The update is verified with a SELECT query.

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 3000,
  3     Stock_Status = 'In Stock',
  4     Last_Updated = TO_DATE('20-09-2026','DD-MM-YYYY')
  5 WHERE Inventory_ID = 103;
```

1 row updated.

```
SQL>
SQL> COMMIT;
```

Commit complete.

9. Inventory Status Reports

9.1 Complete Inventory Report

Combines the Inventory, Supplier and Product tables to show the current stock position of every item (after the update in Section 8).

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Supplier_Name,
  4     p.Product_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status,
  7     i.Last_Updated
  8 FROM Inventory i
  9 JOIN Supplier s
10 ON i.Supplier_ID = s.Supplier_ID
11 JOIN Product p
12 ON i.Product_ID = p.Product_ID
13 ORDER BY i.Inventory_ID;
```

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
101	Shenzhen SoundWave Electronics	JBL Earbuds	5000	In Stock	19-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
102	Guangzhou AudioTech Co.	Bluetooth Speaker	2500	In Stock	19-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
103	Dongguan BassPro Trading	Wireless Headphones	3000	In Stock	20-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
--------------	---------------	--------------	----------------	--------------	-----------

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
104	Huizhou EarPod Manufacturing	Cotton Shirt	1500	In Stock	19-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
105	Foshan Wireless Audio Ltd.	Denim Jeans	3500	In Stock	19-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
106	Ningbo SoundTech Ltd.	Python Programming Book	0	Out of Stock	19-SEP-26

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
--------------	---------------	--------------	----------------	--------------	-----------

```

INVENTORY_ID
-----
SUPPLIER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS      LAST_UPDA
-----
      107
Shanghai AudioPro Industries
DBMS Fundamentals                                4200
In Stock      19-SEP-26

INVENTORY_ID
-----
SUPPLIER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS      LAST_UPDA
-----
      108
Xiamen Wireless Hub
Mixer Grinder                                1800
In Stock      19-SEP-26

INVENTORY_ID
-----
SUPPLIER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS      LAST_UPDA
-----

```

```

INVENTORY_ID
-----
SUPPLIER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS      LAST_UPDA
-----
      109
Wenzhou Smart Audio Co.
Electric Kettle                                2750
In Stock      19-SEP-26

INVENTORY_ID
-----
SUPPLIER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS      LAST_UPDA
-----
      110
Hangzhou Bluetooth Tech
Football                                0
Out of Stock      19-SEP-26

10 rows selected.

```

9.2 Total Stock by Supplier

The total stock quantity held by each supplier is calculated using SUM with GROUP BY.

```
SQL> SELECT
  2     s.Supplier_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Supplier s
  5 JOIN Inventory i
  6 ON s.Supplier_ID = i.Supplier_ID
  7 GROUP BY s.Supplier_Name
  8 ORDER BY s.Supplier_Name;
```

```
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
Dongguan BassPro Trading
      3000

Foshan Wireless Audio Ltd.
      3500

Guangzhou AudioTech Co.
      2500
```

```
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
Hangzhou Bluetooth Tech
      0

Huizhou EarPod Manufacturing
      1500

Ningbo SoundTech Ltd.
      0
```

```
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
Shanghai AudioPro Industries
      4200

Shenzhen SoundWave Electronics
      5000

Wenzhou Smart Audio Co.
      2750
```

```
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
Xiamen Wireless Hub
      1800
```

9.3 Product with Zero Stock

Products whose stock quantity is zero are listed along with the supplier responsible for restocking them.

```
SQL> SELECT
2     i.Product_ID,
3     s.Supplier_Name,
4     i.Stock_Quantity,
5     i.Stock_Status
6 FROM Inventory i
7 JOIN Supplier s
8 ON i.Supplier_ID = s.Supplier_ID
9 WHERE i.Stock_Quantity = 0;

PRODUCT_ID
-----
SUPPLIER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
106
Ningbo SoundTech Ltd.
0 Out of Stock

110
Hangzhou Bluetooth Tech
0 Out of Stock

PRODUCT_ID
-----
SUPPLIER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
```

9.4 Available and Unavailable Product Count

The number of inventory records in each stock status is counted using GROUP BY.

```
SQL> SELECT
2     Stock_Status,
3     COUNT(*) AS Total_Product
4 FROM Inventory
5 GROUP BY Stock_Status
6 ORDER BY Stock_Status;

STOCK_STATUS      TOTAL_PRODUCT
-----
In Stock          8
Out of Stock      2

SQL> SELECT
```

9.5 Low Stock Report

Products that are still available but have fewer than 2,000 units left are flagged so that buyers and suppliers can plan restocking.

```
SQL> SELECT
2     Stock_Status,
3     COUNT(*) AS Total_Product
4 FROM Inventory
5 GROUP BY Stock_Status
6 ORDER BY Stock_Status;
```

```
STOCK_STATUS      TOTAL_PRODUCT
-----
In Stock          8
Out of Stock      2
```

```
SQL> SELECT
2     i.Product_ID,
3     s.Supplier_Name,
4     i.Stock_Quantity
5 FROM Inventory i
6 JOIN Supplier s
7 ON i.Supplier_ID = s.Supplier_ID
8 WHERE i.Stock_Quantity > 0
9 AND i.Stock_Quantity < 2000
10 ORDER BY i.Stock_Quantity;
```

```
PRODUCT_ID
-----
SUPPLIER_NAME
-----
STOCK_QUANTITY
-----
104
Huizhou EarPod Manufacturing
1500
108
```

```
108
Xiamen Wireless Hub
1800
```

```
PRODUCT_ID
-----
SUPPLIER_NAME
-----
STOCK_QUANTITY
-----
```

2 rows selected.

10. Conclusion

The Supplier and Inventory tables were successfully created and connected with the existing Product table using primary and foreign keys. Supplier and product information can be maintained along with stock quantities and stock status. Available and unavailable products can be tracked using SQL queries. Inventory reports such as complete inventory, supplier-wise stock, availability, and low-stock reports can also be generated for the Alibaba B2B wholesale sourcing platform.